EMERGENT COORDINATION IN MULTI-AGENT LANGUAGE MODELS

Christoph Riedl
D'Amore-McKim School of Business,
Khoury College of Computer Sciences, and
Network Science Institute
Northeastern University
Boston, MA 02115, USA
c.riedl@northeastern.edu

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ABSTRACT

When are multi-agent LLM systems merely a collection of individual agents versus an integrated collective with higher-order structure? We introduce an information-theoretic framework to test—in a purely data-driven way—whether multi-agent systems show signs of higher-order structure. This information decomposition lets us measure whether dynamical emergence is present in multi-agent LLM systems, localize it, and distinguish spurious temporal coupling from performance-relevant cross-agent synergy. We implement a practical criterion and an emergence capacity criterion operationalized as partial information decomposition of time-delayed mutual information (TDMI). We apply our framework to experiments using a simple guessing game without direct agent communication and minimal group-level feedback with three randomized interventions. Groups in the control condition exhibit strong temporal synergy but little coordinated alignment across agents. Assigning a persona to each agent introduces stable identity-linked differentiation. Combining personas with an instruction to "think about what other agents might do" shows identity-linked differentiation and goal-directed complementarity across agents. Taken together, our framework establishes that multi-agent LLM systems can be steered with prompt design from mere aggregates to higher-order collectives. Our results are robust across emergence measures and entropy estimators, and not explained by coordination-free baselines or temporal dynamics alone. Without attributing human-like cognition to the agents, the patterns of interaction we observe mirror well-established principles of collective intelligence in human groups: effective performance requires both alignment on shared objectives and complementary contributions across members.

1 INTRODUCTION

Recent advances in generative AI (specifically LLMs) have led to tremendous advances in multi-agent systems (Qu et al., 2024; Hong et al., 2024; Qian et al., 2023; Li et al., 2024a; Subramaniam et al., 2025). Multi-agent systems often show impressive performance increases over single-agent solutions (Wu et al., 2023; Chen et al., 2023; Li et al., 2024b; Tao et al., 2024). A key argument behind such performance gains are claims to "greater-than-the-sum-of-its-parts" effects from differentiated agents (Chen et al., 2023, p.1). Connecting multiple differentiated agents has the potential benefit to leverage unique contributions that would not be available if the task was assigned to a single agent (Fazelpour & De-Arteaga, 2022; Tollefsen et al., 2013; Luppi et al., 2024; Lix et al., 2022). Conceptually, any group gains depend on synergy and emergence (Theiner, 2018; Riedl et al., 2021; Page, 2008; Hayek, 1945).[1] Despite impressive performance of many multi-agent systems we do not yet have a principled understanding when and how such synergy arises, what role agent differentiation plays, and how to steer it systematically. This all points to a crucial need for deeper understanding of collective intelligence in multi-agent systems, specifically whether multi-agent

[1]This is true for both human groups and multi-agent systems.

LLMs exhibit any synergy or complementarity at all—the necessary prerequisite for any absolute team-over-solo gains.

Determining whether multi-agent systems function as genuine collectives—rather than merely aggregating agents—requires a principled measure of synergy. We follow a purely data-driven approach to assess whether these systems exhibit higher-order synergy, characterized by structural coupling and joint information about future states and task outcomes, as evidence of emergent collective behavior. Intuitively, synergy refers to information about a target that a collection of variables provide only jointly but not individually (Rosas et al., 2020; Humphreys, 1997).[2] We develop a principled framework (Figure 1a) building on new insights in information theory (Rosas et al., 2020; Mediano et al., 2022b). Our goal is to develop methods to measure emergent role specialization, determine whether multi-agent systems show such signs of emergent synergy, explore whether synergy enables increased performance, and explore whether emergent behavior can be systematically steered with prompting.[3]

We apply our framework to study multi-agent systems of GPT-4.1, LLAMA-3.1-8B, LLAMA-3.1-70B, GEMINI 2.0 FLASH, and QWEN3 agents solving a simple group guessing task (Figure 1b) with three treatment interventions: a control condition, a condition that assigns a persona to each agent, and one that uses personas with an instruction to "think about what other agents might do" (a theory of mind (ToM) prompt). The paper is organized into addressing three concrete research questions:

RQ1: Do multi-agent LLM systems possess the capacity for emergence?

RQ2: What functional advantages—such as synergistic coordination and higher performance—arise when multi-agent systems exhibit emergence?

RQ3: Can we design prompts, roles, and reasoning structures that steer the internal coordination regimes of multi-agent to encourage outcome-relevant, goal-directed synergy?

Our findings provide evidence for multi-agent LLM system capacity for emergence and that it underpins performance. Through a variety of complementary analyses and robustness tests, we show that coordination style across interventions is very different. While emergence is present in all, only the ToM-prompt condition leads to groups with identity-linked differentiation and goal-directed complementarity across agents: they operate as a dynamically stable, integrated, goal-directed unit. This is consistent with research on human groups showing that performance gains are not automatic (e.g., Stasser & Titus, 1985) and that complementarity needs to be balanced with integration and goal alignment (Theiner, 2018; DeChurch & Mesmer-Magnus, 2010; Riedl et al., 2021).

This paper helps us understand when and how multi-agent systems exhibit higher-order properties, their internal coordination structure, and how to control them. This can inform multi-agent system design by showing how to combine agents effectively. We make four contributions:

- Novel framework to quantify emergent properties in multi-agent systems based on information decomposition including conditional/residual variants and outcome-relevant partial information decomposition.
- Principled diagnostic approaches to localize where synergy resides and distinguish it from alternative explanations (such as heterogeneous learning rates). Specifically, we develop two surrogate null distribution tests (row-shuffle to probe identity-locked structure; column-shuffle for dynamic alignment). This design lets us tease apart "good" synergy aligned with task goals from spurious or misaligned synergy.
- Demonstrate how to steer emergence with specific prompts. Prompt-level manipulations causally change higher-order dependencies and reliably induces distinct coordination regimes, shifting collectives from spurious and misdirected synergy to stable and goal-aligned complementarity driven by differentiated identities.
- Internal coordination is measurable and controllable with interventions. Groups differ in variance, stability, and adaptability—properties relevant to reliability and deployment.

[2]The key claim explored in this paper is conditional, cross-agent synergy—i.e., synergistic information and coordinated differentiation across agents given the multi-agent constraint (not absolute outperformance over a solo agent). We do not attempt to establish team-over-solo superiority on this task.

[3]See GitHub for replication code `https://github.com/riedlc/AI-GBS`.

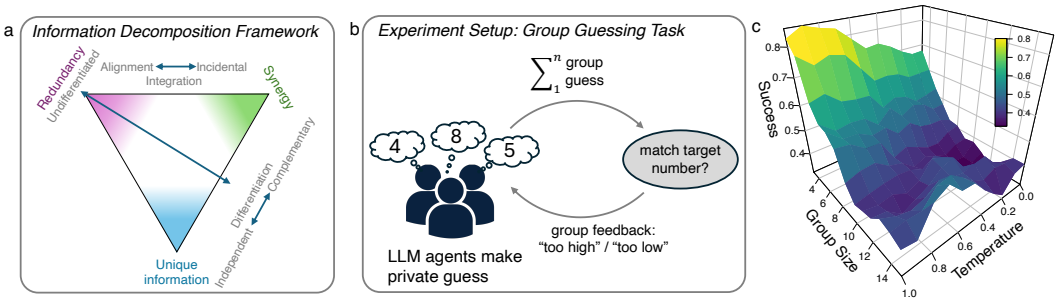

Figure 1: **a)** Information decomposition provides framework to explain tension in multi-agent systems. Agents are either undifferentiated or differentiated, provide independent or complementary information, which is either well aligned or incidental (adapted from Luppi et al., 2024). **b)** Experiment setup of the group binary search task. **c)** Preliminary experiments testing different group sizes and temperature settings. Surface values were smoothed using a local 3×3 weighted averaging filter, giving higher weight to each cell's original value to reduce noise while preserving local structure.

2 METHOD

Group Task. We study a group guessing game (developed by Goldstone et al. (2024) who called it "group binary search") without communication: agents propose integers whose sum needs to match a randomly generated hidden target number. Agents are unaware of each others' guesses and the size of their group, and receive only group-level feedback "too high" or "too low." The task is challenging because identical strategies induce oscillation and only complementary strategies yield success. This setting naturally pits redundancy (alignment) against synergy (useful diversity), and admits clear nulls via row-wise (break identities) and column-wise (break alignment) surrogates. The binary search setting is a minimalist testbed to isolate emergence of cross-agent complementarity and dynamic coordination, not to establish team-over-solo superiority. The implementation of a textbook-like XOR dynamic allows us to study the benefits of emergence capacity conditional on a multi-agent setup (and not to establish team-over-solo superiority). Research on humans has shown that groups can reliably solve this task through the emergence of specialized, complementary roles (Goldstone et al., 2024).

Analytical Framework. How would we know if a multi-agent system shows emergent properties that could suggest that the sum is more than its parts? We develop a principled test based on a recent formal theory of dynamic emergence based on information decomposition (Rosas et al., 2020; Mediano et al., 2022b; Bedau & Humphreys, 2008). Our framework connects emergence with information about a system's temporal evolution—future states of the whole—with information that cannot be traced to the current state of its parts. This framework is based on partial information decomposition (PID; Williams & Beer, 2010) and time-delayed mutual information (TDMI; Luppi et al., 2022), is data-driven, quantifiable, and amenable to empirical testing with falsifiable conjectures against specific null hypotheses. Combining this with permutation tests, focused either on breaking connections between agents or connections across time allow us not only to quantity emergent properties but also to localize them in the system. This is crucial for multi-agent systems because we care about distinguishing synergy among specialized, differentiated agents from mere dynamic alignment.

We implement three tests. The first, termed *emergence capacity*, captures ability of the multi-agent system to host *any* synergy. It measures the predictive synergy from two agents' current states to their *joint* future state, quantified using mutual information $I(\cdot;\cdot)$. For each pair of agents (i, j) and time t, let the sources be $X_{i,t}$ and $X_{j,t}$, and define the bivariate target as the joint future state $t+\ell$, $T_{ij,t+\ell} \equiv (X_{i,t+\ell}, X_{j,t+\ell})$, where ℓ is the integration timescale (Mediano et al., 2022a). From the joint contingency table over $(X_{i,t}, X_{j,t}, T_{ij,t+\ell})$ we compute a two-source PID of the predictive information

$$I(\{X_{i,t}, X_{j,t}\}; T_{ij,t+\ell}) = \mathrm{UI}_i + \mathrm{UI}_j + \mathrm{Red}_{ij} + \mathrm{Syn}_{ij} \tag{1}$$